

SSPA Specialty - Geometry

Questions Sprint for Full Score

T4 area + T5 volume/geometry · Two geometric traps total attack · 75 minutes · 1-to-3 Online Lesson

Corresponding textbooks: Scored test paper 1 + paper 2 geometry questions (accounting for about 25-35% of the paper) · Area · Volume · Combined figures

Core traps: □ T4 area formula trap · T5 geometry/volume trap — ● SSPA compulsory exam

SSPA association: ● High frequency The average score loss rate for geometry questions is 40%, formula confusion is the number one killer

Prerequisite knowledge: L3-L6 area (parallelogram · triangle · trapezoid · polygon)

Class goal: ① Area formula muscle memory ② Zero confusion in volume calculations ③ Combined graph segmentation strategies ④ Correct unit conversion

Student Name:

Class:

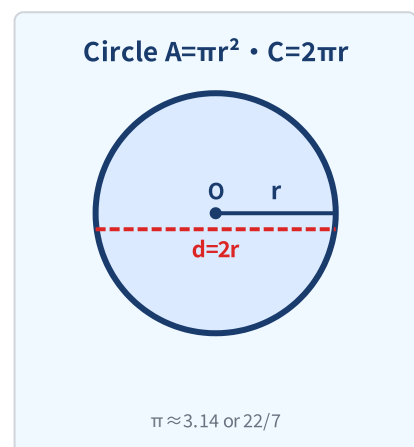
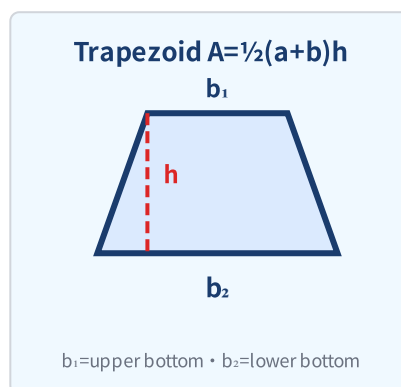
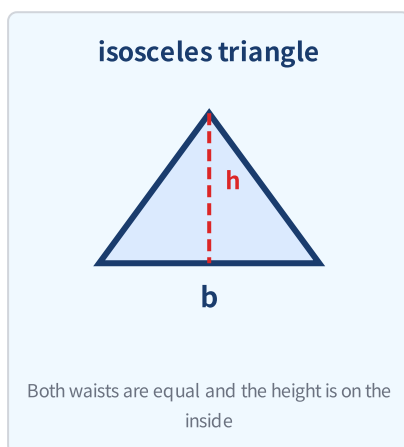
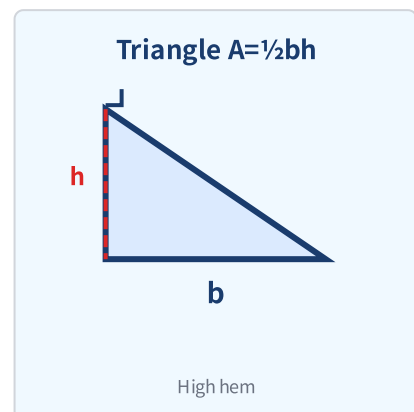
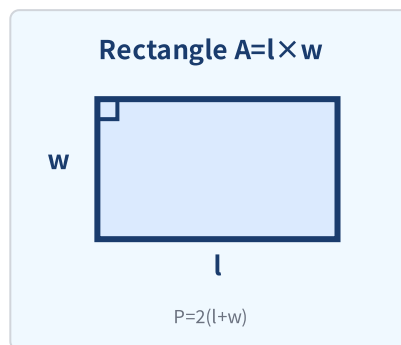
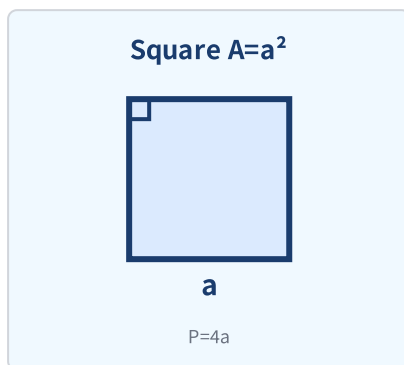
Date:

Time Spent:

I. Geometry warm-up (total 5 questions, 5 minutes)

#	Question	Difficulty	Working Space
W1	Write the formula for the area of a parallelogram.		
W2	Write the formula for the area of a triangle.		
W3	Write the formula for the area of a trapezoid.		
W4	The rectangle is 8 cm long and 5 cm wide. Area = ? Perimeter = ?		
W5	The side length of the square is 6 cm. Area = ? Perimeter = ?		

Quick reference for geometric figures (this Lesson has/have Question total use)



 **The most common mistakes in SSPA geometry questions:** ① Confusing the area formula and the perimeter formula ② Forgetting $\div 2$ (triangle/trapezoid) ③ The height must be the vertical distance ④ Wrong writing of the unit (cm^2 vs cm)

II. Core Knowledge + Worked Examples

Knowledge Point 1: Area Trap Total Attack (T4 Area Formula) ● SSPA required test

Five major area formulas (Must have muscle memory!):

- ① Area of square = side length $\times$ side length
- ② Area of rectangle = length $\times$ width
- ③ Area of parallelogram = base $\times$ **Height** (Height is the vertical distance, not the hypotenuse!)
- ④ Area of triangle = base $\times$ height $\div 2$ ($\div 2$ is easiest to forget!)
- ⑤ Area of trapezoid = **(upper base + lower base) $\times$ height $\div 2$** (not "add height and multiply by 2")

Unit trap: The area unit is cm^2/m^2 , not cm/m ! The answer must be written in square units!

SVG 1: Ladder annotation diagram

Figure 1: Trapezoid area = (upper base + lower base) $\times$ height $\div 2$



Example KP1-1: Trapezoid area formula trap

The upper base of the trapezoid is 6 cm, the lower base is 10 cm, and the height is 4 cm.
Area = ?

✗ Common mistakes

$$(6+10+4) \div 2 = 10 \text{ cm}^2$$

Add more height! The formula is (upper bottom + lower bottom) $\times$ height $\div 2$

✓ Correct solution

$$(6+10) \times 4 \div 2 = 32 \text{ cm}^2$$

First add the two bases = 16, $\times$ height = 64, $\div 2 = 32$

Example KP1-2: Height of parallelogram vs hypotenuse

The base of a parallelogram is 8 cm, the height is 5 cm, and the hypotenuse is 6 cm.
Area = ?

✗ Common mistakes

$$8 \times 6 = 48 \text{ cm}^2$$

Used a bevel! Area of parallelogram = base $\times$ height (vertical distance)

✓ Correct solution

$$8 \times 5 = 40 \text{ cm}^2$$

Height = vertical distance = 5 cm, not hypotenuse 6 cm

💡 **Tip: "Multiply the height of a flat base by multiplying it, divide a triangle by two, add two bases of a trapezoid, multiply the height by dividing by two, and use the hypotenuse as a trap."**

⚠ The three most common mistakes in T4: (1) Forgetting $\div 2$ for triangles (2) Using the hypotenuse

KP1 Synchronization practice — areacalculate (question 1-15)

#	Question	Difficulty	Working Space
1	The parallelogram has a base of 12 cm and a height of 7 cm. Area = ?		
2	Triangular base 10 cm, height 6 cm. Area = ? (Did you divide by 2?)		
3	The upper base of the trapezoid is 5 cm, the lower base is 9 cm, and the height is 6 cm. Area = ?		
4	The rectangle is 15 cm long and 8 cm wide. Area = ? Perimeter = ?		
5	The side length of the square is 9 cm. Area = ?		
6	The parallelogram has a base of 20 cm, a height of 8 cm, and a hypotenuse of 10 cm. Area = ?		
7	Triangular base 14 cm, height 9 cm. Area = ?		
8	The upper base of the trapezoid is 7 cm, the lower base is 13 cm, and the height is 5 cm. Area = ?		
9	The parallelogram has a base of 3.5 cm and a height of 2 cm. Area = ?		
10	The area of the triangle is 24 cm ² and the base is 8 cm. How high is it?		
11	The area of a trapezoid is 40 cm ² , its upper base is 5 cm, and its lower base is 11 cm. How high is it?		
12	A parallelogram and a triangle have the same base and height. The area of the triangle is 18 cm ² . Area of parallelogram = ?		
13	The base of the triangle is 16 cm and the height is half the base. Area = ?		
14	A parallelogram has an area of 72 cm ² and a height of 8 cm. bottom = ? (Note: not the hypotenuse!)		
15	Two identical trapezoids form a parallelogram. Each trapezoid has an upper base of 4 cm, a lower base of 10 cm, and a height of 6 cm. The area of the parallelogram = ?		

Knowledge Point 2: Volume Trap Total Attack (T5 Geometry/Volume) SSPA required test

Core formulas and traps of volume:

- ① Volume of cuboid = length $\times$ width $\times$ height

- ② Volume of cube = side length $\times$ side length $\times$ side length (cube of side length)
- ③ **Unit trap |||SEP|||: The volume unit is cm^3/m^3 (cubic), not cm^2 (square)! Capacity conversion**
- ④ **Surface area vs volume confusion |||SEP|||: Surface area = sum of the areas of all faces (unit cm^2), volume = length $\times$ width $\times$ height (unit cm^3) : $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$**
- ⑤ **Surface area vs volume confusion:** Surface area = sum of the areas of all faces (unit cm^2), volume = length $\times$ width $\times$ height (unit cm^3)

SVG 2: Cuboid stereogram

Figure 2: Volume of cuboid = length $\times$ width $\times$ height (unit: cm^3)



Example KP2-1: Volume vs surface area confusion

A cuboid is 5 cm long, 4 cm wide and 3 cm high. Find (a) volume (b) surface area.

✗ Common mistakes

$$\text{Volume} = \text{surface area} = 5 \times 4 = 20$$

Confusing volume and surface area with wrong units

✓ Correct solution

$$\text{Volume} = 5 \times 4 \times 3 = 60 \text{ cm}^3$$

Surface area =

$$2(5 \times 4 + 5 \times 3 + 4 \times 3)$$

$$= 2(20 + 15 + 12) = 94 \text{ cm}^2$$

The volume is cubic ($\times 3$ times), and the surface area is the sum of the areas of the six sides

Example KP2-2: Capacity conversion trap

A water tank is 20 cm long, 15 cm wide and 10 cm high. How many liters of water can be filled?

✗ Common mistakes

$$20 \times 15 \times 10 = 3000 \text{ L}$$

$$3000 \text{ cm}^3 \neq 3000 \text{ L} ! 1 \text{ L} = 1000 \text{ cm}^3$$

✓ Correct solution

$$\text{Volume} = 20 \times 15 \times 10 = 3000 \text{ cm}^3$$

$$= 3000 \div 1000 = 3 \text{ L}$$

cm^3 to L to $\div 1000$

⚠ T5 must remember: $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$. The unit of volume is "cubic" not "square"!

KP2 synchronization practice — volume and table area (question 16-28)

#	Question	Difficulty	Working Space
16	The cuboid is 6 cm long, 4 cm wide and 2 cm high. Volume = ?		

17	The side length of the cube is 5 cm. Volume = ?		
18	The cuboid is 10 cm long, 5 cm wide and 3 cm high. (a) Volume = ? (b) Surface area = ?		
19	The inside of the tank is 30 cm long, 20 cm wide, and 15 cm deep. Volume of water = ? How many liters?		
20	The surface area of the cube is 150 cm ² . Side length = ? Volume = ?		
21	The volume of a cuboid is 240 cm ³ , its length is 10 cm and its width is 6 cm. High = ?		
22	A fish tank is 40 cm long, 25 cm wide, and 30 cm high. How many liters of water can be filled? (1 L = 1000 cm ³)		
23	The side length of the cube is 8 cm. Surface area = ? Volume = ? (Note that the units are different!)		
24	The sum of the lengths of all sides of a cuboid is 48 cm. 6 cm long, 4 cm wide. High = ? Volume = ?		
25	The volume of a cube is 512 cm ³ . Side length = ? Surface area = ?		
26	The original water in the water tank is 5 L. After a stone is placed, the water level rises and the volume becomes 5.8 L. What is the volume of the stone in cm ³ ?		
27	The length of a cuboid is 8 cm, its width is half its length, and its height is twice its width. Volume = ?		
28	Cut a cube with side length 10 cm into 8 identical small cubes. Volume of each small cube = ? Side length = ?		

Knowledge Point 3: Combination Graphics Strategy (T4 + T5 Comprehensive) ●

SSPA Required Exam

Three-step method for solving combined graphics problems:

- ① **Divide |||SEP|||: Split the combined graphics into basic graphics (rectangle, triangle, trapezoid, parallelogram)** Mark the dimensions |||SEP|||: Mark the required base, height, length, and width for each basic graphics (note that some dimensions need to be calculated)
- ② **Calculate separately and then merge:** Add (merge graphics), subtract (hollow out graphics)
- ③ **Volume combination:** Divide into multiple cuboids/cubes first, find the volumes separately and then add them up

Overlap trap: When two graphics overlap, do not repeat calculations!

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Figure 3: Combined shape = rectangle + triangle (division method)



Example question KP3-1: Area of combined graphics (refer to Figure 3)

Find the total area of the combined shapes in Figure 3. Rectangle 12×8 , triangle base 10 and height 6.

✗ Common mistakes

$$12 \times 8 + 10 \times 6 = 96 + 60 = 156 \text{ cm}^2$$

Triangle Forgot $\div 2$!

✓ Correct solution

$$A = 12 \times 8 = 96 \text{ cm}^2$$

$$B = 10 \times 6 \div 2 = 30 \text{ cm}^2$$

$$\text{Total} = 126 \text{ cm}^2$$

Use the correct formula for each basic graph

Example KP3-2: Combined volume (superposition of two cuboids)

An L-shaped object consists of two cuboids: a lower cuboid $10 \times 5 \times 4$ cm and an upper cuboid $6 \times 5 \times 3$ cm stacked at the lower end. Total volume = ?

✗ Common mistakes

$$10 \times 5 \times 4 + 6 \times 5 \times 3 = 200 + 90 = 290 \text{ cm}^3$$

It seems correct, but if the two cuboids overlap, they need to be subtracted.

✓ Correct solution

$$V_1 = 10 \times 5 \times 4 = 200 \text{ cm}^3$$

$$V_2 = 6 \times 5 \times 3 = 90 \text{ cm}^3$$

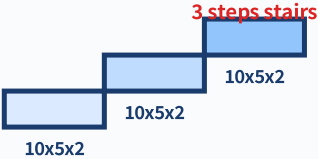
$$\text{Total} = 290 \text{ cm}^3 \text{ (without overlap)}$$

Calculate the volumes separately and add them after confirming there is no overlap.

⚠ The most frequent errors in combined graphics: (1) Forgetting $\div 2$ for triangles (2) Forgetting to calculate the "hidden size" (3) Repeating calculations for overlapping parts

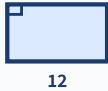
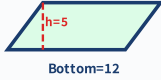
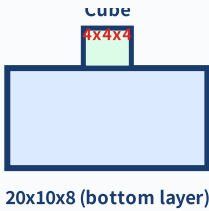
KP3 synchronization practice — compound shape (question 29-40)

#	Question	Difficulty	Working Space
29	A composite figure consists of a rectangle (8×5 cm) and a triangle (base 8 cm, height 4 cm). Total area = ?		
30	Cut out a triangle (base 6 cm, height 4 cm) in a rectangle (10×6 cm). Remaining area = ?		

31	A trapezoid (upper base 4 cm, lower base 10 cm, height 6 cm) is topped by a triangle (base 10 cm, height 3 cm). Total area = ?		
32	Two identical parallelograms (base 8 cm, height 5 cm) are put together to overlap a triangle (base 8 cm, height 2.5 cm). Total area = ?		
33	An L-shaped plane consists of two rectangles: 12×4 cm vertically and 8×5 cm horizontally. They overlap a 4×4 cm square. Total area = ?		
34	 A stair-shaped volume consists of 3 levels. Each level is 10 cm long, 5 cm wide and 2 cm high. Three levels are stacked to form a ladder. Total volume = ?		
35	Cut a trapezoid (upper base 4 cm, lower base 8 cm, height 5 cm) out of a rectangle (15×10 cm). Remaining area = ?		
36	A small cuboid ($4 \times 3 \times 2$ cm) is dug out of a large cuboid ($10 \times 8 \times 6$ cm). Remaining volume = ?		
37	A square has a side of 10 cm and is cut diagonally into two triangles. Area of each triangle = ?		
38	Two identical trapezoids (upper base 3 cm, lower base 7 cm, height 4 cm) are assembled into a hexagon "bottom to bottom". Total area = ?		
39	A rectangular park $50 \text{ m} \times 30 \text{ m}$. There is a straight path 2 m wide in the middle that runs through the long sides. Usable grass area = ?		
40	The cube has a side length of 6 cm. The surface is painted and then cut into 27 small cubes with a side length of 2 cm. How many small cubes are there with "paint on one side"?		

III. Geometry Ultimate Sprint (question 41-45 • champion level challenge)

#	Question	Difficulty	Working Space
41	A hexagon can be divided into a rectangle (8×6 cm) and two identical triangles (each base 4 cm and height 6 cm). Area of hexagon = ?		

42	If the side length of a cube is doubled, how many times its volume will increase? How many times does the surface area change?		
43	Rectangle  12 Parallelogram  h=5 Bottom=12 A rectangle and a parallelogram have equal bases and equal heights (base 12 cm, height 5 cm). What is the difference between the areas of the two shapes? Why?		
44	 20x10x8 (bottom layer) A combined solid: a cuboid ($20 \times 10 \times 8$ cm) below and a cube (side length 4 cm) above in the center. Total volume = ? Surface area = ? (Note: The joint surface does not count as surface area)		
45	Draw an irregular shape on graph paper (1 cm^2 per square). There are 24 complete squares, and about 8 half squares and most half squares. Estimated area = ? Write out your estimation method.		

IV.areaformula cheat sheet

graphics	areaformula	mostCommon Mistake
square	side length $\times$ side length	confused with perimeter
rectangle	length $\times$ width	confused with perimeter
parallelogram	Bottom $\times$ high	Use hypotenuse as height
triangle	bottom $\times$ height $\div 2$	forget $\div 2$
trapezoid	(upper bottom + lower bottom) $\times$ height $\div 2$	Add height / forget $\div 2$

V. The Lessoncorecommon errorssummary

 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

#	common error	trap	Correct Approach
1	Forget the area of a triangle $\div 2$	T4	Remember: base $\times$ height is a parallelogram, and a triangle only needs half
2	Parallelogram uses hypotenuse	T4	The height must be the vertical distance, and the hypotenuse must be the interference item
3	Add height to the trapezoidal formula	T4	The formula is (upper + lower) $\times$ height $\div 2$, not (upper + lower + height) $\times 2$
4	Area vs perimeter confusion	T4	Perimeter = sum of side lengths (cm), area = calculated by formula (cm ²)
5	Volume vs surface area confusion	T5	Volume = cm ³ (space occupied), surface area = cm ² (sum of six side areas)
6	Capacity conversion error	T5	1 cm ³ = 1 mL, 1000 cm ³ = 1 L, must $\div 1000$
7	Combined graphics overlap and repeat calculations	T4/T5	After segmentation, check whether there is overlap. If there is overlap, you must subtract it once.
8	Area/volume unit spelled incorrectly	T4/T5	The area must be written in cm ² /m ² and the volume must be written in cm ³ /m ³
9	Hidden dimensions are not calculated	T4/T5	Some dimensions are not given directly and need to be inferred from other known dimensions.

Strategies for getting perfect marks on geometry questions in sub-tests **Must read**

Three-step verification method for geometry questions:

- ① **Formula check** |||SEP|||: Write the formula $\rightarrow$ Substitute the numbers $\rightarrow$ Confirm $\div 2$ / Not $\div 2$ Unit check
- ② : Area $\rightarrow$ cm²/m², volume $\rightarrow$ cm³/m³, length $\rightarrow$ cm/m Reasonability check
- ③ : For example, the area of a triangle < the area of a parallelogram with the same base and the same height? Is the area of the trapezoid reasonable?: For example, the area of a triangle < the area of a parallelogram with the same base and the same height? Is the area of the trapezoid reasonable?

 **Tips for perfect score in geometry: "Recite formulas for area, divide triangles by two, use height instead of slant for squares, volume is cubic, and units must be understood clearly."**